# Transfer: gravity and motion

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Apply gravity, projectile and orbital reasoning to a new set of cases.

**SUCCESS CRITERIA - I can...**

1. Choose a diagram or equation. State what it treats as true before calculating.
2. Connect diagrams to forces and motion.
3. Check units and explain the limits of conclusions.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “model” and “gravity” to explain your answer. Sentence starters are optional.

**model**: a drawing, explanation or equation for a physical situation

**gravity**: an attractive interaction between masses

**projectile**: an object moving after launch under the stated forces

## 01 Warm-up: bring back an earlier idea

A satellite follows a circular orbit at a constant **5 km/s**.
1. **Draw velocity** and acceleration at the top and side of the orbit.
2. **Explain the different directions**.

**Word help:** orbit = a path around another body under gravity

**Sentence starter:** Both … . They differ in … because … .

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** $F_g = mg$; $a_c = v^2/r$; $h = \frac{1}{2}gt^2$; range = $v_x t$.
Match each equation to its case: near-surface weight, circular motion, or horizontal launch. Use $g$ = **10 m/s²**.

**Worked reasoning:** A toy horizontal projectile falls for **2 s** while moving at **4 m/s**. Range is **8 m**. The vertical motion sets time; the horizontal motion sets range once time is known.

**Your turn:** Draw two coordinated timelines, one for each direction, with a shared **2 s** endpoint.

**Sentence starter:** In my work, I used … to show … . I labeled it … .



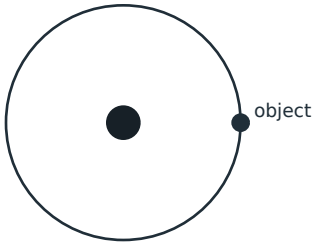
Add velocity / acceleration arrows.

**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

A satellite moves in a circular orbit.
1. **Draw velocity** and gravity arrows.
2. **Explain why floating** does not mean zero gravity.

**Sentence starter:** I think … . I can point to …, which shows …, so … .




The center is the attracting body; the small dot on the circle is the orbiting object.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 04 Apply the physics

A ball launches horizontally at **6 m/s**. Supplied fall time is **2 s**. **Find range**. If height stays same and launch speed halves, find new range. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

**Sentence starter:** I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

**Marzano check - circle:** 1 2 3

## 05 Explain your reasoning

A Sun-orbit returns every **2 years**. Why can period alone not prove an Earth encounter? **Name one measurement or observation** you would request.

**Sentence starter:** I think … . I can point to …, which shows …, so … .

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

Two fixed masses attract with **144 N** at separation r. **Find the force** at separation **6** r and **explain the factor**. **Label the masses**, field or force relevant to the question.

**Word help:** separation = distance between the specified objects or centers

**Sentence starter:** I think … . I can point to …, which shows …, so … .



Label separation if needed.

Source Object

**Marzano check - circle:** 1 2 3

## 07 Exit check

Surface field is **8 N/kg**. At center distance **2** R, find field and force on a **3 kg** object. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Separate horizontal and vertical motion**

Label x and y; state positive directions.

Velocity:

Acceleration / force:

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

Design two launches from the same height with ranges in a **1**: **3** ratio. **State what you change** and what remains fixed. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

**Sentence starter:** My design includes … . It does what is needed by … .



Add velocity / acceleration arrows.

**Marzano check - circle:** 1 2 3